Fixed $m_G L$ campaign

Theory, methods and preliminary results

Current status and scope

The canonical analysis contains 41 physical ensembles and 168 independent chains: 1,648,000 measured sweeps, 1,556,340 after independent thermalization cuts, and 1,547,750 used in complete blocks. There are 41 numerical mass estimates and 10 usable preliminary coupling interpolations. All three targets, $c = 0.5, 1, 2$, have $N_t = 2, 3, 4$; $c = 1$ also has $N_t = 6$. Every new bootstrap operation uses exactly 500 replicas. No new Monte Carlo simulations are launched by this reanalysis. Raw histories remain on qbig.

The register now also includes three $N_t = 1$ matched-coupling ensembles: twelve independent 8,000-sweep chains completed on lnode19 A100 GPUs on 14 October. Their couplings are the three current $N_t = 2$ tuning estimates. For these rows, the folder's c_{ref} denotes that source target; the actual $N_t = 1$ mass is measured without imposing $m_G N_t = c_{\mathrm{ref}}$. They are excluded from fixed-c interpolation and continuum fits.

The four 80,000-sweep chains at $c = 0.5, N_t = 3, N_s = 80, \beta = 4.4$ have the explicit flag `length_outlier`. Their length differs from the 8,000-sweep standard; the flag does not establish a physics bias. They are pooled with four independent 8,000-sweep chains of identical parameters. Eight 10,000-sweep chains are explicitly labelled prior inputs from the fixed-β/N_t campaign, at $N_t = 2, \beta = 3.0$ and 2.1.

All defined estimates remain available for visual review. Flags identify window drift, incomplete error plateaus, limited block counts, chain differences, numerical covariance problems and interpolation sensitivity. They propagate to preliminary continuum results. The editable `analysis/mg_nt_fixed/review.csv` records manual window choices and inclusion decisions. No flag is silently converted into a rejection.

How to read the results

Each ensemble lives at `analysis/mg_nt_fixed/c_<c>/Nt_<Nt>/Ns_<Ns>/beta_<beta>/`. It contains correlators, all fit windows, chain statistics and compact bootstrap arrays. Its `plots/` directory shows correlator fits, residuals and the full mass-window scan; `plots/first_stage/` shows every chain's thermalization cut, bootstrap error versus block size, the selected block and block counts. Blocking examples are kept outside this PDF. Comparisons sit at the nearest common physics level. The CSV column guide is `docs/mg_nt_fixed/csv_columns.md`.

Theory: what is being held fixed?

The model is three-dimensional SU(2) Yang–Mills theory with Wilson action

$$S_W = -\frac{\beta}{2} \sum_p \mathrm{Re}\,\mathrm{Tr}\, U_p + \text{constant}.$$

Both transverse directions are periodic. Tangential links on two temporal Dirichlet surfaces are fixed to the identity [1]. The separation is $L = aN_t$; the transverse box size is $L_s = aN_s$. Current simulations store $N_t + 1$ spatial-link slices and N_t temporal-link layers, with no closing layer. Thus N_t counts links between the fixed surfaces. Historical wrapping-sector outputs retain their original executable provenance. With fixed surface links their action factorizes into the physical slab and an independent auxiliary sector, so removing that sector preserves physical slab expectation values.

For integer transverse separation r, define

$$G(\mathbf{x}) = \prod_{t=0}^{N_t-1} U_t(\mathbf{x}, t),$$
$$C_G(r) = \left\langle \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}[G(\mathbf{x})G^\dagger(\mathbf{x} + r\hat{e})] \right\rangle.$$

Origins and both transverse axes are averaged. The normalization gives $C_G(0) = 1$. The residual boundary transformations act as $G \mapsto V_L G V_R^\dagger$; in $GG^\dagger$, the V_R factors cancel and the trace removes the remaining V_L conjugation. This is the $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ symmetry of the principal chiral model (PCM). At $N_t = 1$ the slab plaquettes reduce exactly to nearest-neighbor PCM interactions [1].

The asymptotic decay of C_G measures the lightest transverse screening scale with overlap in this holonomy channel, under single-state dominance. With $\widehat{m} = am_G$, the trajectory is

$$c = m_G L = \widehat{m} N_t \in \{0.5, 1, 2\}, \qquad \xi_{\mathrm{exp}}/a = 1/\widehat{m} = N_t/c, \qquad \xi_{\mathrm{exp}}/L = 1/c.$$

The correlation length is twice, equal to, or half the slab thickness. These are geometry-to-correlation-length ratios; identifying a phase or a limiting effective theory requires additional evidence. The smallest target has only a factor-two length separation. A controlled PCM long-distance interpretation would benefit from $m_G L \ll 1$ and agreement of further dimensionless observables. The present scope is the holonomy mass.

Increasing N_t at fixed c gives, step by step,

$$a/L = 1/N_t \to 0, \qquad \widehat{m} = c/N_t \to 0, \qquad \xi_{\mathrm{exp}}/a = N_t/c \to \infty.$$

Thus the same physical thickness is increasingly well resolved in lattice units, using the measured screening length as the ruler. The transverse volume is separately controlled through $m_G L_s = \widehat{m} N_s$.

From correlator to continuum mass

At each N_t, nearby couplings are simulated and their masses determine $\beta_\star$ satisfying $\widehat{m}(\beta_\star, N_t)N_t = c$. The tuning condition fixes $m_G L$ by construction. The nontrivial continuum information is the coupling required to satisfy it. In the isotropic Wilson convention $\beta = 4/(ag_{3,0}^2)$ [2],

$$g_{3,0}^2 L = (ag_{3,0}^2)N_t = 4N_t/\beta_\star,$$
$$Q(N_t) \equiv m_G/g_{3,0}^2 = \frac{am_G}{ag_{3,0}^2} = \frac{\widehat{m}\beta_\star}{4} = \frac{c\beta_\star}{4N_t}.$$

The plots fit $Q(N_t) = Q_0 + d/N_t$ only ($p = 1$). The equivalent $\beta_\star/N_t$ plots are in each target's comparison directory. At finite spacing these are bare-coupling coordinates; interpretation in continuum coupling units assumes the usual continuum matching. At fixed boundary prescription and large transverse volume, the result determines the local inverse relation between $m_G L$ and $g_3^2 L$.

The $p = 1$ form is a restricted cutoff ansatz. Fits using all resolutions and omitting the coarsest resolution show sensitivity to this choice. Two points determine a line with zero residual degrees of freedom and provide no residual test of the ansatz. Statistical bands are conditional on the chosen mass windows, interpolation windows and volumes. The CSV also reports a separate conservative linear-response bound on the continuum intercept from independently signed two-point interpolation shifts at each resolution. This bound is not added in quadrature to the statistical error.

Mass analysis and correlated uncertainty

Each chain is cut independently with MSER-5 and a 500-sweep minimum; the largest cut across monitored observables is used. Profiles at cadence five enter as equilibration monitors. Integrated autocorrelation times use the self-consistent Sokal window and are reported in sweeps [3]. One composite sweep is a heatbath update and five overrelaxation updates. Holonomy observables are measured every sweep. The second-moment length is monitored using ensemble structure factors,

$$\xi_{2\mathrm{nd}}/a = \frac{1}{2\sin(\pi/N_s)}\sqrt{\frac{\widetilde{C}_G(0)}{\widetilde{C}_G(p_{\min})} - 1}, \qquad ap_{\min} = 2\pi/N_s.$$

The same joint replicas propagate its error. Ratios are formed after ensemble averaging; unresolved replicas are recorded explicitly.

Complete nonoverlapping blocks are formed inside each chain. Candidate lengths scan the whole post-cut history, starting at five sweeps. The chosen length exceeds twice the largest measured autocorrelation time and uses the earliest common three-length error plateau when available. Each curve point is a separate 500-replica bootstrap SE. Plateau tolerances are 5% or two approximate SE uncertainties, whichever is larger; the latter use Welch–Satterthwaite degrees of freedom and the Monte Carlo SE of 500 draws. This is a diagnostic stability criterion. Missing plateaus are flagged. Fewer than 30 complete blocks is a warning; 50 is the target.

Pooling, fitting and window review

If chain k contributes n_k complete blocks of B_k measurements, $w_k = n_k B_k / \sum_j n_j B_j$. In each of exactly 500 replicas, draw n_k block vectors with replacement within chain k, then form $C^{(b)} = \sum_k w_k \overline{C}_k^{(b)}$ [4]. The draw acts on the entire separation vector. Chain identities and weights are fixed. The ensemble mean is $\sum_k w_k \overline{C}_k$. Independent chains retain their own cuts, autocorrelations and block boundaries. Split-history and between-chain discrepancies are reported separately; maxima over many correlated observables are review diagnostics.

The asymptotic two-image model is [1, Eq. (4.14)]

$$C_G(r) \simeq A\{K_0(\widehat{m}r) + K_0[\widehat{m}(N_s - r)]\}.$$

Its interpretation requires a stable long-distance window and finite-volume control. Additional images, excited contributions and lattice dispersion may matter outside that regime. Each central and bootstrap fit minimizes the diagonal-error objective. Parameter errors are the SD of the same fit over the 500 vector replicas. The complete covariance is used to report $\chi^2_{\text{full}} = r^T \,\text{Cov}^{-1}\, r$ after fitting. This statistic is descriptive: the fitted parameters do not minimize it, so a nominal analytic chi-square p-value is not appropriate. No p-value is computed and no chi-square gate is imposed. Singular covariance leaves this diagnostic undefined while retaining a defined diagonal fit.

For the original fixed-c ensembles, all contiguous windows with at least five points are fitted, with every positive $R_{\min}$ and $R_{\max} = N_s/2, N_s/2 - 2, \ldots \geq \lceil N_s/3 \rceil$. The recommendation is the earliest stable 3×3 patch in adjacent endpoints, preferring the largest $R_{\max}$. Pairwise masses must differ by at most two combined bootstrap SEs. Overlapping windows are correlated; this rule is a heuristic for visual review. If no such patch exists, the least-drifting numerical patch is flagged and retained. Failed refit slots remain NaN in the saved 500-slot arrays; success counts and failure reasons are in CSV. Relative error above 25%, low success fraction and a missing mass-error blocking plateau produce flags.

For the new $N_t = 1$ boxes with $N_s = 512, 2048$, the user approved a bounded endpoint scan: seven allowed upper endpoints and logarithmically spaced lower endpoints, followed by dense adjacent checks near three candidate plateaus. Every integer separation inside each window and all 500 refits are retained. The recommendation is the earliest stable patch among tested windows; it does not establish the exhaustive earliest patch. The $N_s = 128$ scan remains exhaustive. A covariance formed from 500 centered replicas has rank at most 499; windows with 500 or more separations therefore have undefined full-covariance χ^2. Fewer independent block vectors can reduce the rank further. No inverse regularization is used to manufacture a goodness-of-fit statistic.

Coupling windows and volume choices

For each c, N_t, all contiguous three-coupling windows are examined using $\log(\widehat{m}N_t) = a_0 + b\beta$. Only centrally bracketed targets yield an interpolation; undefined slopes or unbracketed targets are recorded. Each aligned mass replica gives $\beta_\star^{(b)} = (\log c - a_0^{(b)})/b^{(b)}$. The median and 16–84% interval are quoted. The largest shift relative to a target-bracketing two-point interpolation is a separate systematic. The closest bracketing window is suggested; manual choices remain editable.

The larger $N_s = 96$ boxes are primary at $c = 1, N_t = 4, \beta = 4.95, 5.3$; the $N_s = 48$ versions remain volume diagnostics. At $c = 0.5, N_t = 4, \beta = 5.58394149$, $N_s = 96$ is primary and 128 is a matched-volume check. The $N_t = 4, c = 1$ confirmation ensemble now contributes to tuning. Its earlier independence applies to the historical interpolation only.

Coupling and continuum results

c	N_t	$\beta_\star$	16–84% interval	interp. shift	status
0.5	2	2.9895	[2.9888, 2.9903]	0.00276	flagged
0.5	3	4.3069	[4.3059, 4.3081]	0.000383	flagged
0.5	4	5.5846	[5.5821, 5.5868]	0.000685	flagged
1	2	2.5045	[2.5038, 2.5051]	0.000365	flagged
1	3	3.5985	[3.5968, 3.6002]	0.00201	flagged
1	4	4.6305	[4.6279, 4.6326]	0.00352	flagged
1	6	6.6664	[6.6636, 6.6691]	0.0108	flagged
2	2	1.9624	[1.9603, 1.964]	0.00269	flagged
2	3	2.9428	[2.941, 2.9442]	0.00187	flagged
2	4	3.773	[3.7704, 3.7754]	0.00349	flagged

All entries are preliminary and subject to visual review. The interpolation shift is kept separate from the statistical interval.

c	resolutions	Q_0	boot. SE	interp. bound	dof
0.5	2, 3, 4	0.16331	0.000124	0.000295	1
0.5	3, 4	0.15971	0.000331	0.000134	0
1	2, 3, 4, 6	0.26225	0.000158	0.000647	2
1	3, 4, 6	0.25572	0.000261	0.00108	1
2	2, 3, 4	0.45733	0.000768	0.00163	1
2	3, 4	0.41509	0.00153	0.00268	0

The second row for each target omits its coarsest resolution. Detailed flags, full-covariance residual statistics and bootstrap success counts are in the CSV tables. The chosen primary volumes are not a volume extrapolation.

Resolution dependence is the principal limitation of these extrapolations. Omitting $N_t = 2$ moves Q_0 from approximately 0.16331 to 0.15971 at $c = 0.5$, 0.26225 to 0.25572 at $c = 1$, and 0.45733 to 0.41509 at $c = 2$. These shifts are much larger than the conditional bootstrap errors. They are explicitly flagged using the paired-replica uncertainty of the difference between the two fits. The tables retain all requested preliminary fits; their small statistical errors do not establish a controlled continuum mass. Finer resolutions and visual range selection are the priority before a precision continuum claim.

Why earlier results changed

The original resampler included an outer draw of independent chain identities. The corrected method fixes those identities and resamples blocks within each chain. Earlier bootstrap counts and uncertainty prescriptions are preserved in the checksum-verified qbig archive; they are historical products. This reanalysis additionally pools matching short and long runs with fixed used-measurement weights and applies the requested 500-replica policy to all 38 physical ensembles.

Earlier acceptance also imposed fit p-value and window gates. In particular, the $c = 2, N_t = 3, N_s = 32, \beta = 3.25$ mass was unavailable after those gates, despite numerical windows that can be inspected for a plateau. The new analysis retains numerical windows, flags drift and lets visual review determine their use. The coupling fits previously rejected for local log-linearity are displayed in the comparison directories, including their residuals. Keeping these preliminary results increases coverage; their statistical errors remain conditional on the model and selected window.

Next simulations to consider after review

First inspect flagged mass and coupling windows, especially the $\beta = 3.25$ case and the smallest remaining block counts. Additional independent 8,000-sweep chains can improve precision and between-chain checks without adding another length outlier. At otherwise fixed conditions, doubling effective statistics reduces statistical error by approximately $\sqrt{2}$; halving it requires roughly four times the effective statistics. More sweeps cannot resolve a poor fit model or uncontrolled finite-volume effects. At a fixed block size B, obtaining 50 blocks within every chain requires at least its thermalization cut plus $50B$ sweeps: for a 500-sweep cut this is 10,500 sweeps at $B = 200$ and 12,500 at $B = 240$. Adding more 8,000-sweep chains improves combined statistics but leaves that per-chain count unchanged unless the selected block size changes.

The largest current dataset supports three c values, with four resolutions only at $c = 1$. The next coverage priority is an $N_t = 6$ bracket for $c = 0.5$ and $c = 2$, followed by independent confirmation of the tuned couplings and volume/timing checks. No new simulation is authorized by this report.

New $N_t = 1$ matched-coupling diagnostics

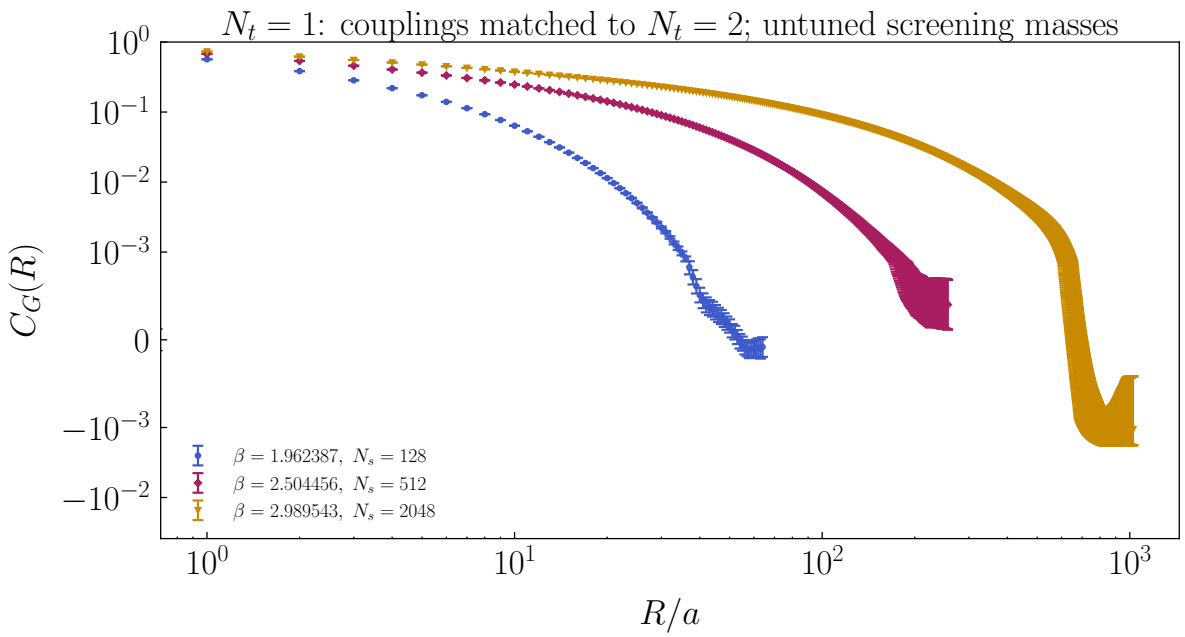
c_{ref}	β	N_s	am_G	boot. SE	$\xi_{2\mathrm{nd}}/a$	min. blocks
0.5	2.989543	2048	0.0067043	3.77e-05	146.7	6
1	2.504456	512	0.028794	6.3e-05	40.59	36
2	1.962387	128	0.14111	0.000219	7.348	37

All twelve chains completed with verified inputs, output counts, boundary conditions and checkpoints, using KLFT revision `f47421d895469b72d4af8a9b8479807f2b09a692` and its native A100 build. Total simulation time was 3.038 A100 GPU-hours. Holonomies were measured every sweep, profiles every five sweeps, and Wilson loops were disabled. The separate P100 validation job 290996 also passed all six CTests, CUDA memory checking and the benchmark; it launched no production.

The transverse sizes 128, 512 and 2048 were explicit candidate input choices. The saved planning records label them unvalidated and contain no quantitative derivation establishing their necessity. At the largest coupling, the first-stage estimate is $\xi_{2\mathrm{nd}}/a \simeq 147 \pm 34$, giving $N_s/\xi_{2\mathrm{nd}} \simeq 14$ at the central value. This is preliminary because independent cuts range from 825 to 3625 sweeps, the maximum estimated autocorrelation time is about 317 sweeps, and $B = 635$ leaves only 6–11 complete blocks per chain. The common blocking plateau is unresolved and split-history/between-chain differences require review. At $N_s = 512$, $\xi_{2\mathrm{nd}}/a \simeq 40.6 \pm 2.9$ and $B = 200$ leaves 36–37 blocks; at $N_s = 128$, $\xi_{2\mathrm{nd}}/a \simeq 7.35 \pm 0.23$ and $B = 200$ leaves 37 blocks. These measured lengths explain why matched couplings cannot be interpreted as the source fixed-c targets at $N_t = 1$.

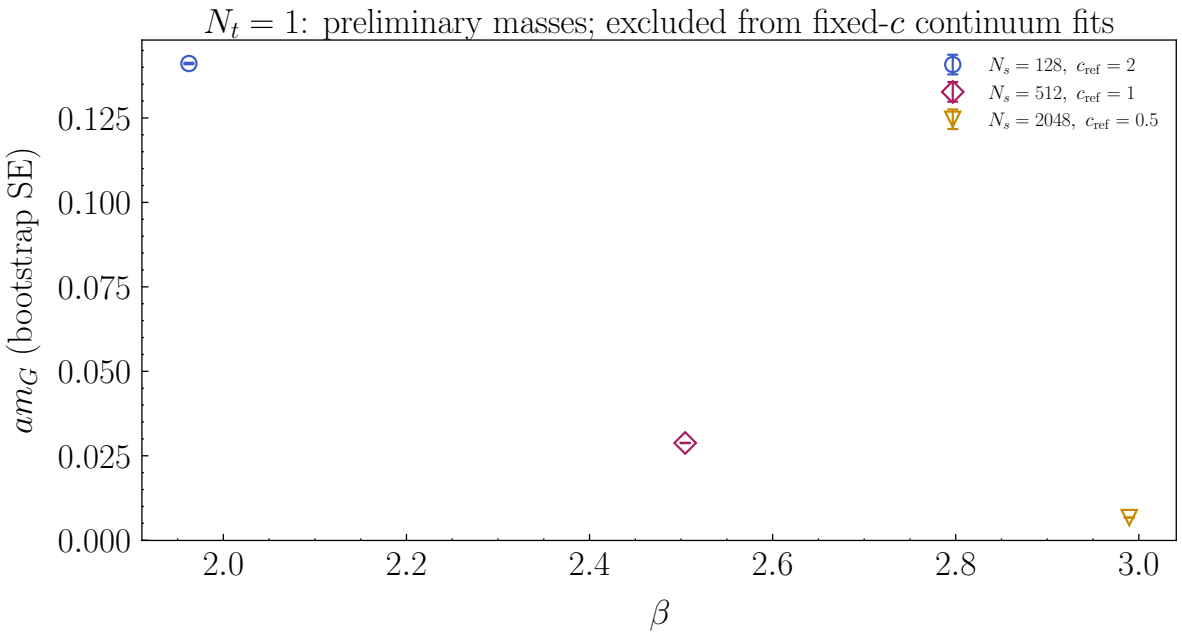
The largest-volume statistics are the immediate concern. At the present block length, 50 complete blocks require 31,750 post-cut sweeps per chain; the most conservative observed cut would bring this to 35,375 total sweeps. This is a planning estimate: autocorrelation and thermalization must be rechecked after extending statistics. A smaller-volume comparison is needed before replacing 2048 by a smaller production size. Adding these untuned points does not extend the fixed-c continuum resolution set.

$N_t = 1$: matched-coupling correlators



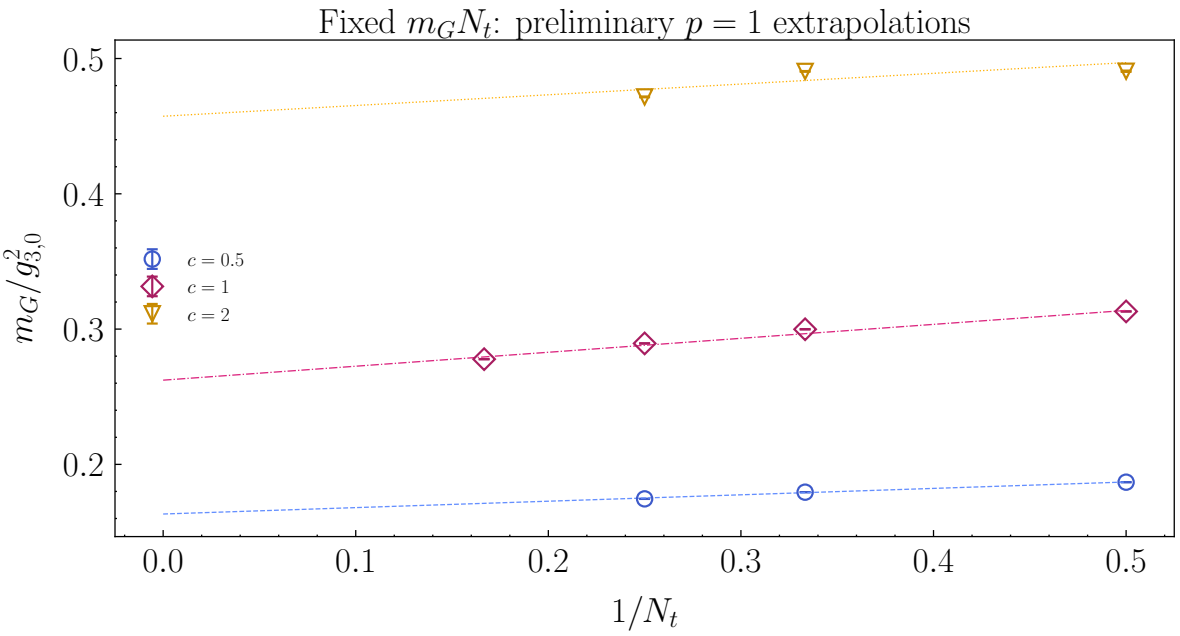
The three volumes correspond to different couplings. Their folder labels identify the source $N_t = 2$ tuning. Error bars use independent-chain, pooled-with-boundaries bootstrap replicas.

$N_t = 1$: preliminary screening masses



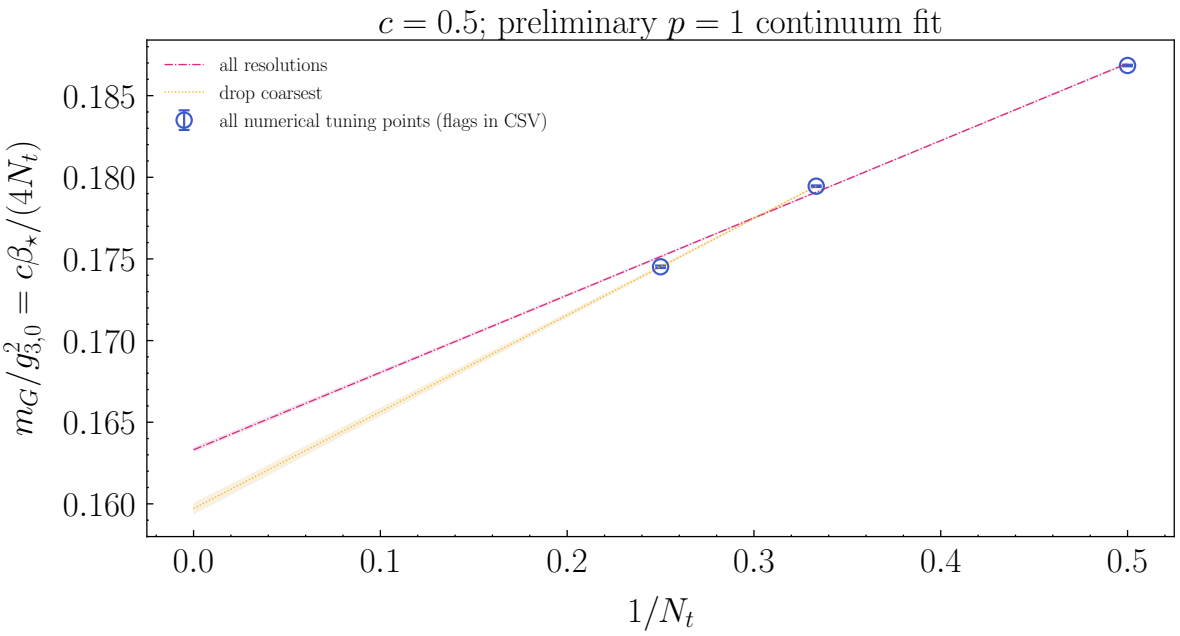
Conditional bootstrap standard errors are shown. Window, blocking, chain-consistency and volume flags remain in the per-ensemble CSVs. These masses are excluded from fixed-c continuum fits.

Continuum comparison: all targets



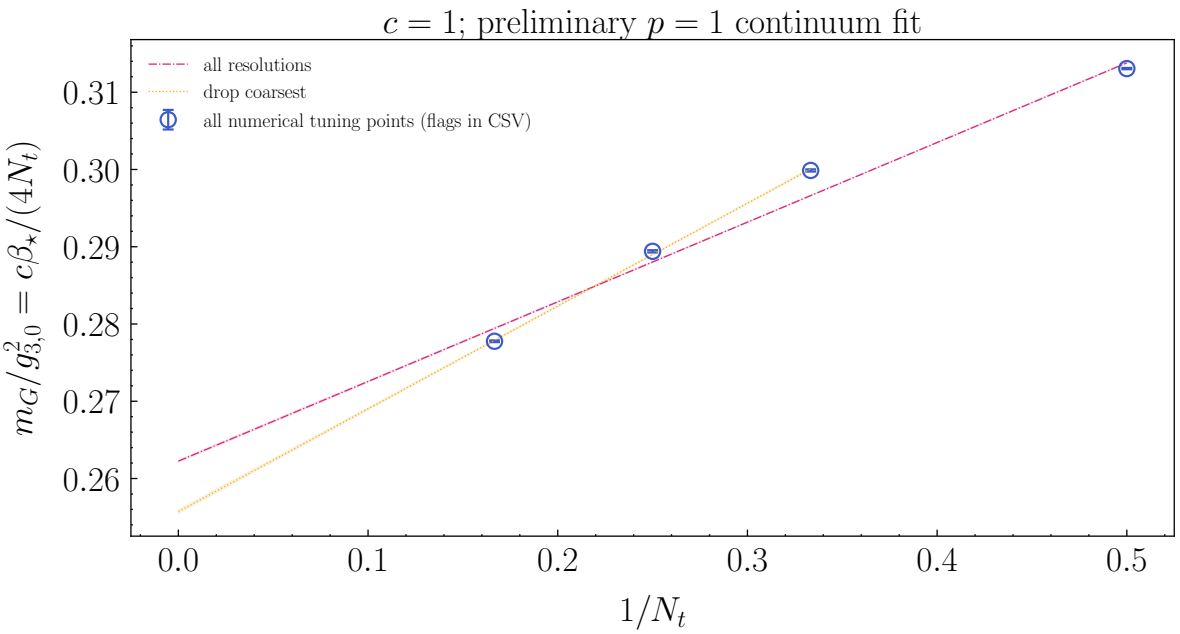
All three targets use the requested $p = 1$ ansatz. Flags remain attached to their input points.

$c = 0.5$: resolution sensitivity



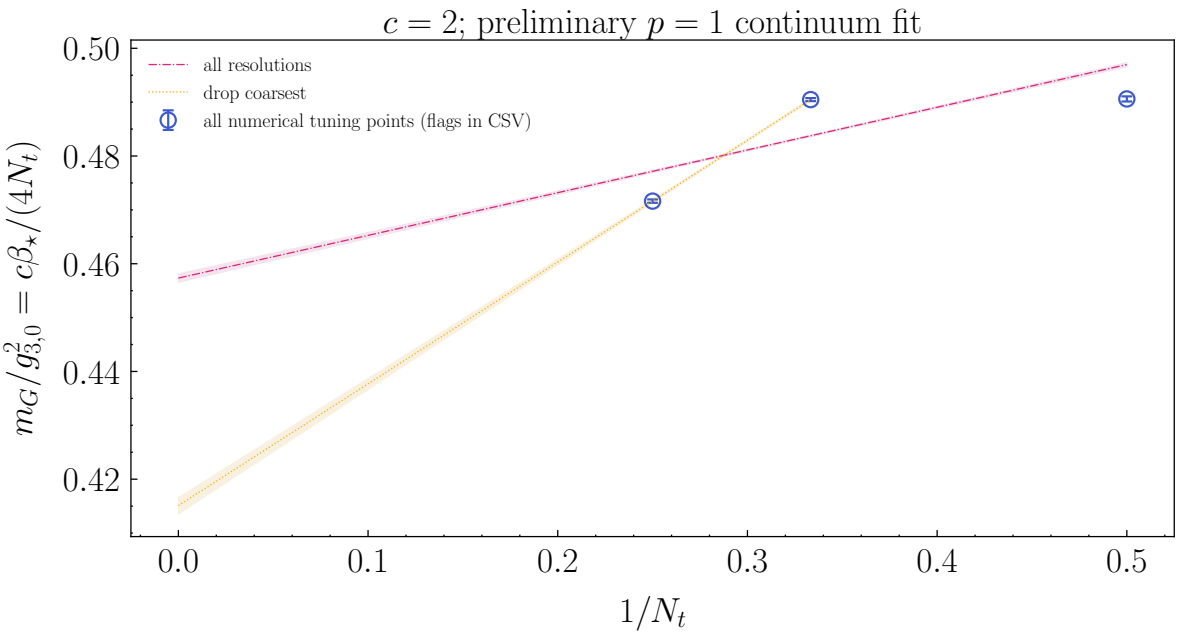
The drop-coarsest line uses two resolutions and has zero residual degrees of freedom. Bands contain conditional statistical uncertainty only.

$c = 1$: four measured resolutions



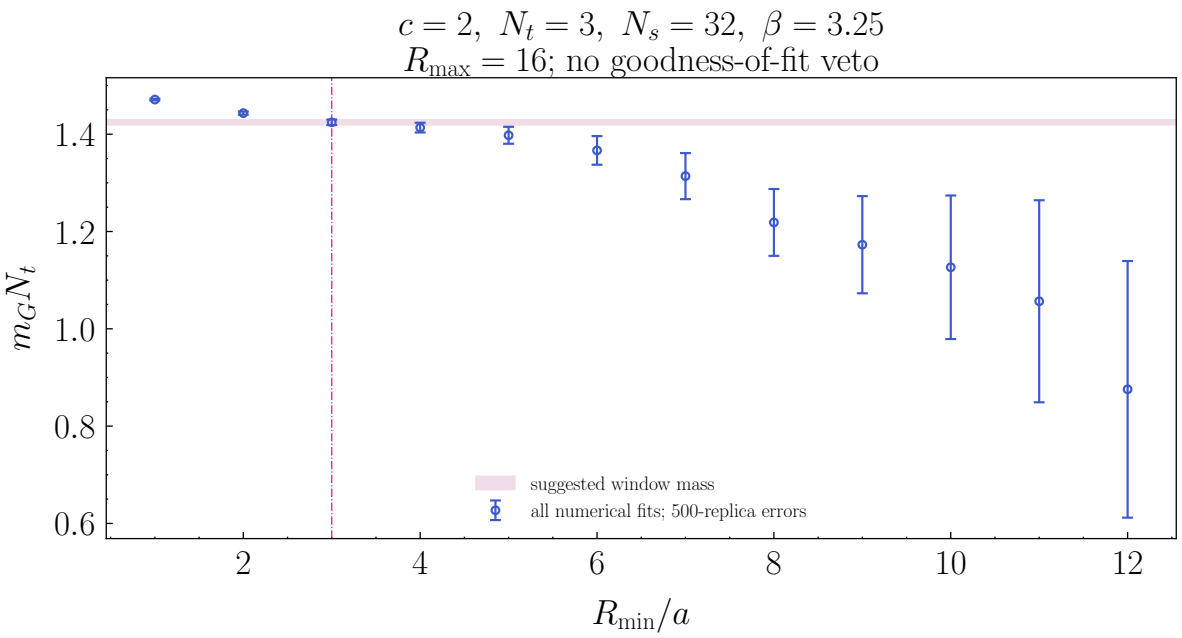
The available $N_t = 6$ bracket extends the continuum comparison. Independent tuning confirmation and further cutoff control remain useful.

$c = 2$: preliminary retained trajectory



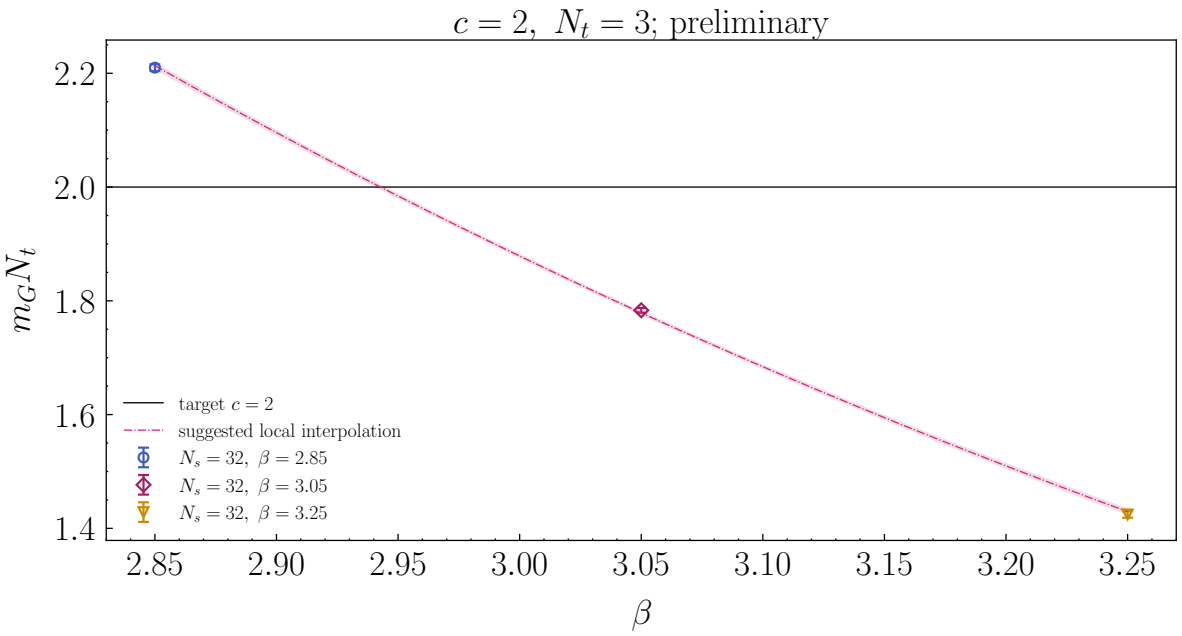
Mass-window and interpolation concerns remain visible in the CSV flags and per-ensemble plots.

Previously gated mass: $N_t = 3, N_s = 32, \beta = 3.25$



Every numerical mass window at this endpoint is shown. The band marks the suggested mass; overlapping-window stability is a diagnostic heuristic. Other endpoints and the correlator residuals are saved alongside this plot.

Example coupling interpolation



The previously gated mass is retained as a preliminary input. All coupling windows and their residual plots are available in the same comparison directory.

Ensemble result register

c	N_t	N_s	β	am_G	SE	window	used sweeps
0.5	2	64	2.9	0.28458	0.000491	3–32	29,760
0.5	2	64	3	0.24513	0.000456	4–32	37,600
0.5	2	64	3.1	0.21427	0.000438	3–32	29,920
0.5	3	80	4.2	0.18478	0.000372	3–40	29,440
0.5	3	80	4.27	0.17315	0.000339	3–40	30,000
0.5	3	80	4.29	0.16951	0.000328	3–40	30,000
0.5	3	80	4.4	0.15159	0.000156	5–40	347,000
0.5	3	80	4.6	0.12704	0.000299	3–40	29,760
0.5	4	96	5.5	0.1334	0.000399	4–48	29,440
0.5	4	96	5.58394149	0.125	0.000371	4–48	29,600
0.5	4	96	5.6	0.12358	0.000333	4–48	29,760
0.5	4	96	5.9	0.10119	0.000325	4–48	30,000
0.5	4	96	6.2	0.082482	0.000326	4–48	29,760
0.5	4	128	5.58394149	0.12562	0.000255	4–64	29,760
1	2	32	2.35	0.6181	0.00139	2–16	29,920
1	2	32	2.5	0.50322	0.000599	1–16	30,000
1	2	32	2.65	0.40906	0.000958	2–16	29,760
1	3	48	3.5	0.36806	0.000895	3–24	30,000
1	3	48	3.7	0.3004	0.000733	3–24	29,920
1	3	48	3.9	0.2481	0.00064	3–24	30,000
1	4	48	4.6	0.25524	0.000661	3–24	29,920
1	4	48	4.61755576	0.25314	0.000718	3–24	30,000
1	4	48	4.95	0.19764	0.000578	3–24	29,760
1	4	48	5.3	0.15342	0.000614	3–24	29,760
1	4	96	4.95	0.19631	0.00042	4–48	30,000
1	4	96	5.3	0.15284	0.000385	4–48	30,000
1	6	96	6.4	0.19015	0.000364	4–48	29,760
1	6	96	6.7	0.16289	0.000486	5–48	30,000
1	6	96	7	0.14187	0.000362	4–48	29,760
2	2	32	1.9	1.0772	0.00403	2–16	29,440
2	2	32	2	0.95897	0.00322	2–16	29,920
2	2	48	2.1	0.83975	0.00392	3–24	37,920
2	3	32	2.85	0.73663	0.00198	2–16	29,760
2	3	32	3.05	0.59445	0.00143	2–16	29,920
2	3	32	3.25	0.47481	0.0019	3–16	30,000
2	4	32	3.7	0.53286	0.00134	2–16	29,920
2	4	32	4	0.40814	0.0016	3–16	30,000
2	4	32	4.3	0.323	0.00125	3–16	29,920

Uncertainties are bootstrap standard errors of am_G. The selected window and used statistics are included so a result can be traced to its CSV and raw-input hashes. The complete chain-level cuts, autocorrelation times, flags and block counts are in `chains.csv` and `autocorrelations.csv`; no chain histories are concatenated.

References

[1] M. Nebbia, *Effects of Dirichlet Boundary Conditions in Yang–Mills Theories*, MSc thesis (2024/2025), local reference `docs/marconebbia_tesi.pdf`.

[2] M. Teper, SU(N) Yang–Mills theory in $2 + 1$ dimensions, Phys. Rev. D 59, 014512 (1999), hep-lat/9804008.

[3] A. D. Sokal, Monte Carlo Methods in Statistical Mechanics: Foundations and New Algorithms, in *Functional Integration*, NATO ASI B 361 (1997), pp. 131–192.

[4] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*, Chapman & Hall (1993), chapters 6 and 8.